When the Tools Don't Fit

Student Success Platforms and the HBCU Retention Gap

Background of Study

- Various student success interventions are needed to ensure optimal graduation rates and positive career and life outcomes for all students navigating their higher education journey, including those at Historically Black Colleges and Universities (HBCUs).

- HBCUs, like other institutions of higher education, invest in student success platforms to structure and coordinate student services. The dominant platforms in this space were designed around the demographic and institutional profiles of Predominantly White Institutions (PWIs).

- No student success platform has been purpose-built for the structural realities of HBCUs, which serve substantially higher proportions of Pell-eligible, first-generation, and basic needs-insecure students while operating with significantly fewer per-student resources. This structural mismatch creates gaps that current platforms are not designed to close.

- This study seeks to identify which institution-level academic and student support indicators are most strongly associated with first-year retention and six-year graduation rates at HBCUs — findings that will inform the design of a cohesive, data-driven student success platform built specifically for the HBCU context.

Colors: **Blue, Gold, Fluorescent Lime Green and Bright Orange accents**

⇒ **Blue** is an academic color used by most educational institutions
⇒ **Gold** shows dominance
⇒ **Fluorescent Lime Green** represents innovation
⇒ **Bright Orange** represents enthusiasm

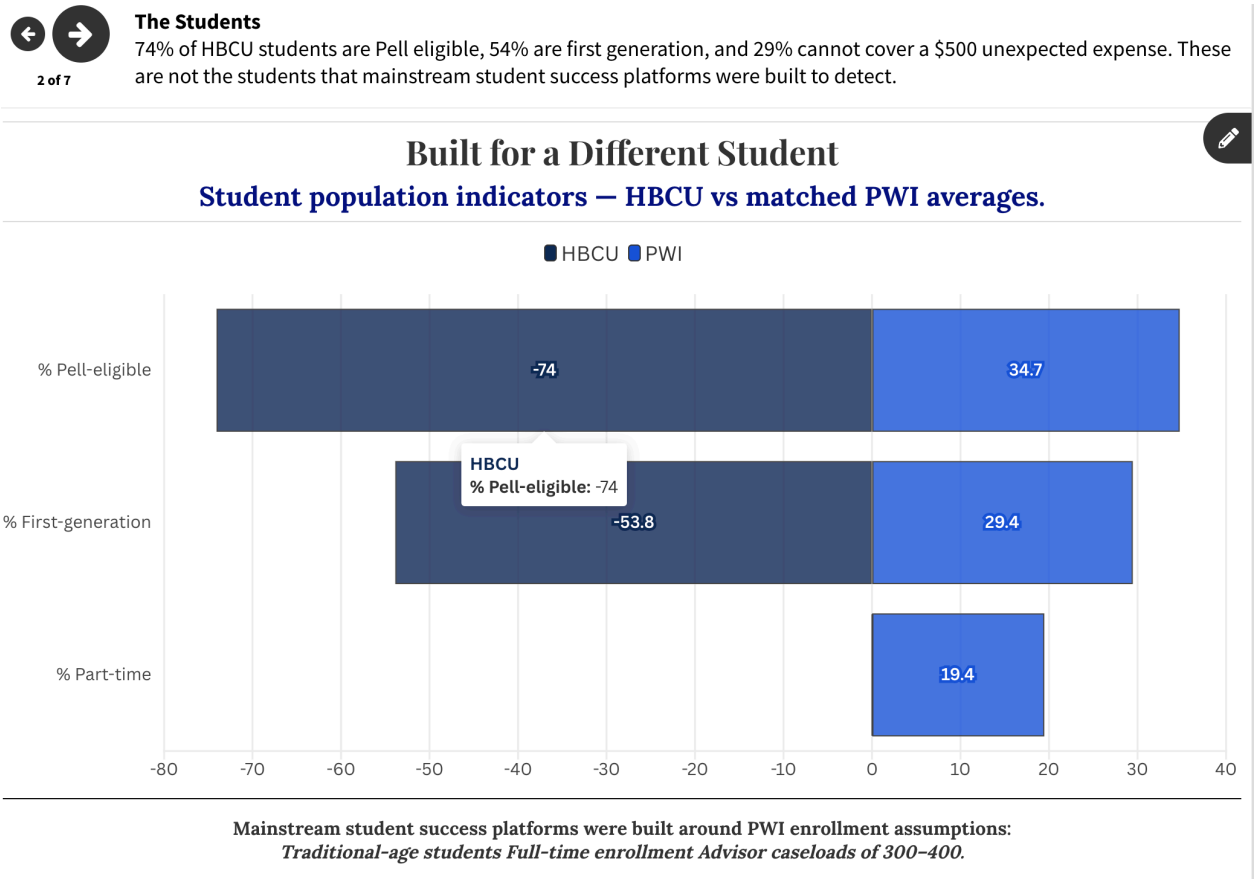
Research Questions

1) **Do student success platforms designed for PWIs help HBCUs close gaps in retention and graduation rates?**
2) **And if not, what type of student success platform would be more effective for HBCUs?**

Deeper Inquiry

1) **Which institution-level academic and student support indicators correlate most strongly with first-year retention and six-year graduation rates at HBCUs?**
2) **Do HBCUs and PWIs differ significantly on key student success outcomes — and if so, does the pattern suggest that platform adoption alone cannot close the gap?**
3) **What structural data quality issues (multicollinearity, imbalance, missing data) must be diagnosed and remediated before building predictive ML models?**

Slide 1 — The Student Transition Slide



Today, I want to begin with the students at the center of this story.

At HBCUs, student success work is not only about academic performance. It is also about financial insecurity, first-generation navigation, family obligations, and institutional resource constraints.

In this dataset, **74% of HBCU students are Pell eligible**, **54% are first-generation**, and **29% report that they cannot cover a $500 unexpected expense**.

These numbers matter because they show that many HBCU students experience risk differently. Their challenges are not always captured by traditional academic indicators like GPA, credit completion, or course attendance alone.

And that is the central problem: **these are not the students mainstream student success platforms were originally built to detect**.

Slide 2 — The Gap Bar Chart

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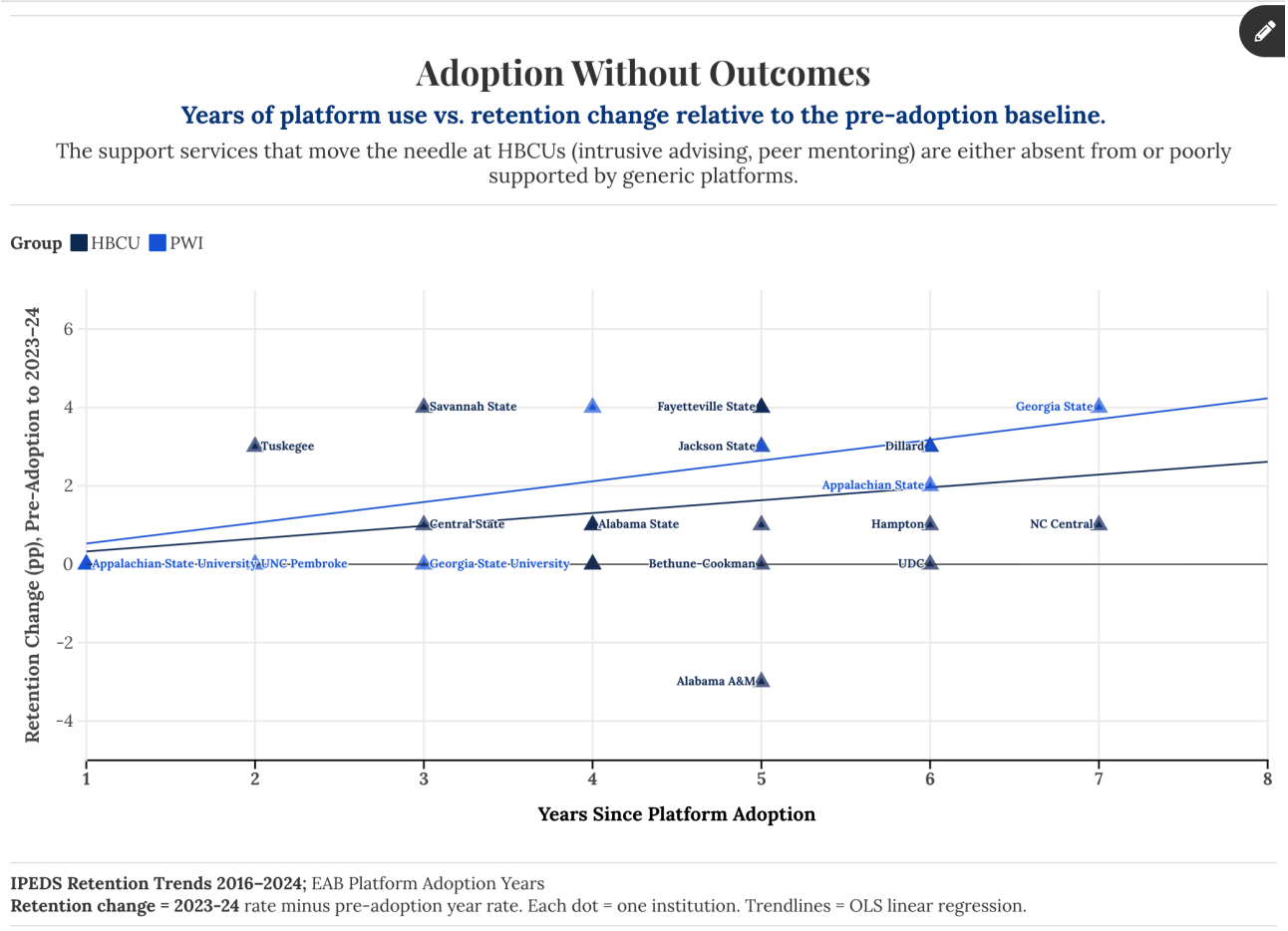
The Gap
Between 2017 and 2022, the retention gap between HBCUs and matched PWIs held steady at 10.5 to 10.6 percentage points. Eight years of platform adoption produced no measurable change.



This slide shows the retention gap between HBCUs and matched PWIs from 2017 to 2022.

What stands out is not just the existence of the gap, but its stability.

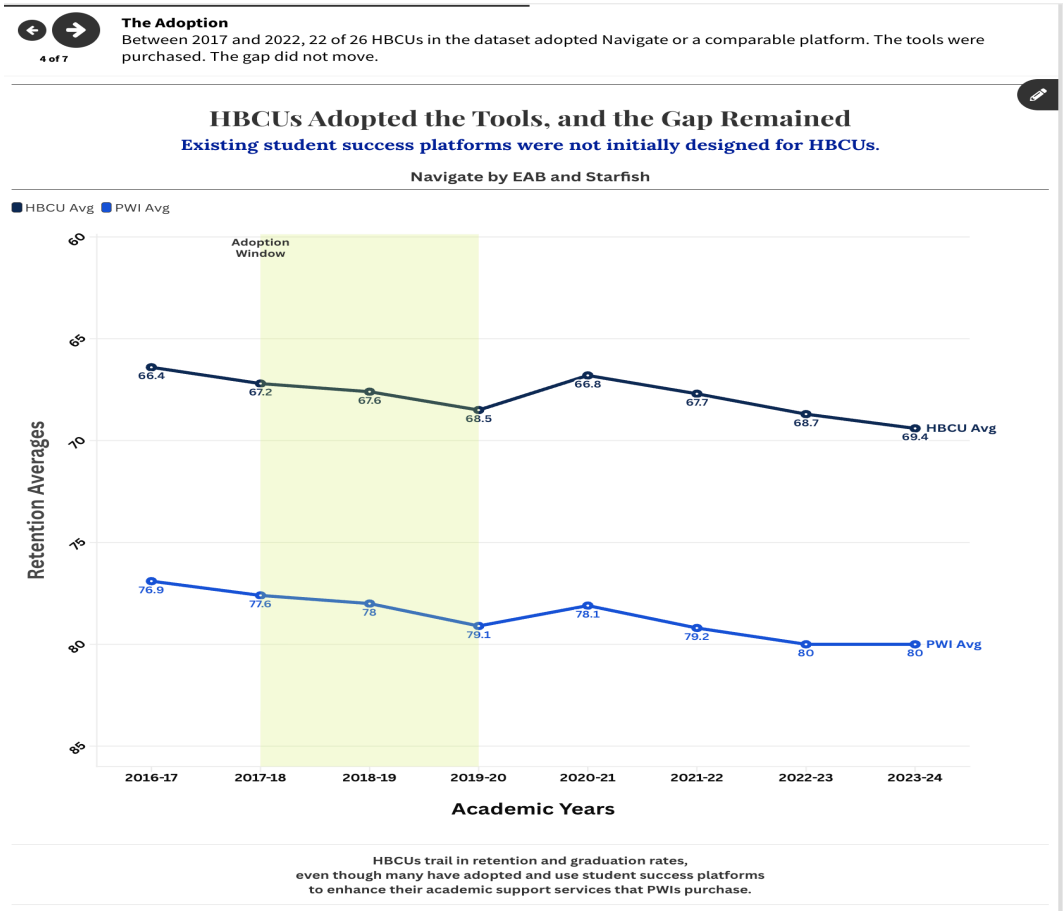
Across this period, the gap held at roughly **10.5 to 10.6 percentage points**.

That means that even as institutions adopted student success technologies, the difference in retention outcomes between HBCUs and comparable PWIs did not meaningfully change.

In other words, **eight years of platform adoption produced no measurable movement in the retention gap**.

That raises an important question: if the tools were implemented, why did the outcomes remain the same?

Slide 3 — The Adoption



This slide helps answer part of that question.

Between 2017 and 2022, **22 of the 26 HBCUs** in the dataset adopted Navigate or a comparable student success platform.

So, the issue is not that HBCUs ignored technology. The issue is not that HBCUs failed to modernize. The issue is not that HBCUs were unwilling to invest in student success systems.

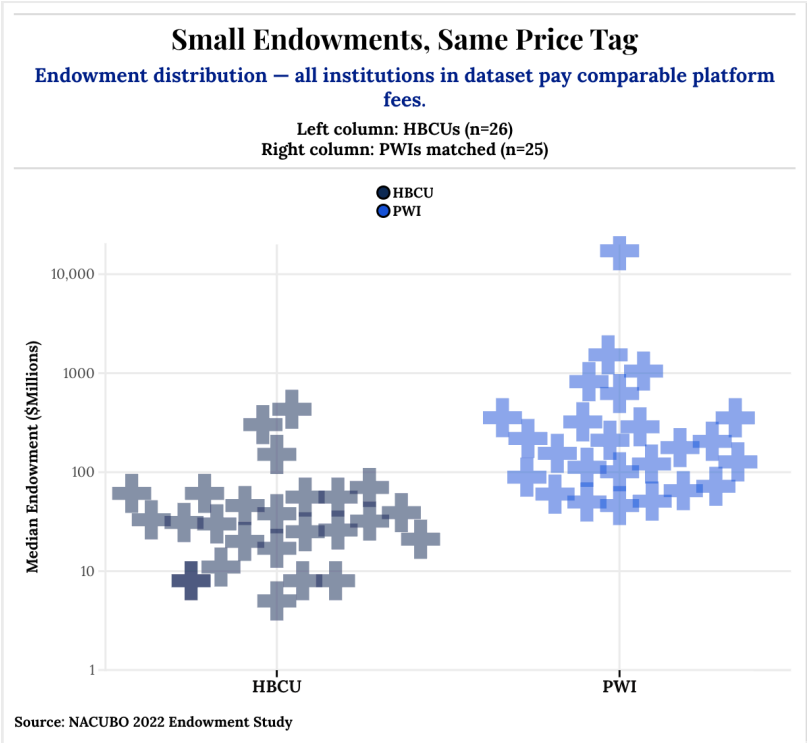
The tools were purchased.

The platforms were adopted.

But the gap did not move.

This suggests that adoption alone is not enough. It also suggests that the design of the tool matters, especially when the student population has needs that are more complex, more relational, and more resource-dependent than the platform was designed to address.

Slide 4 — The Financial Reality



This slide shows the financial reality behind the retention gap.

The average HBCU endowment in this dataset is **$53.6 million**.

The average PWI endowment is **$26.7 billion**.

That is a **498-to-1 ratio**.

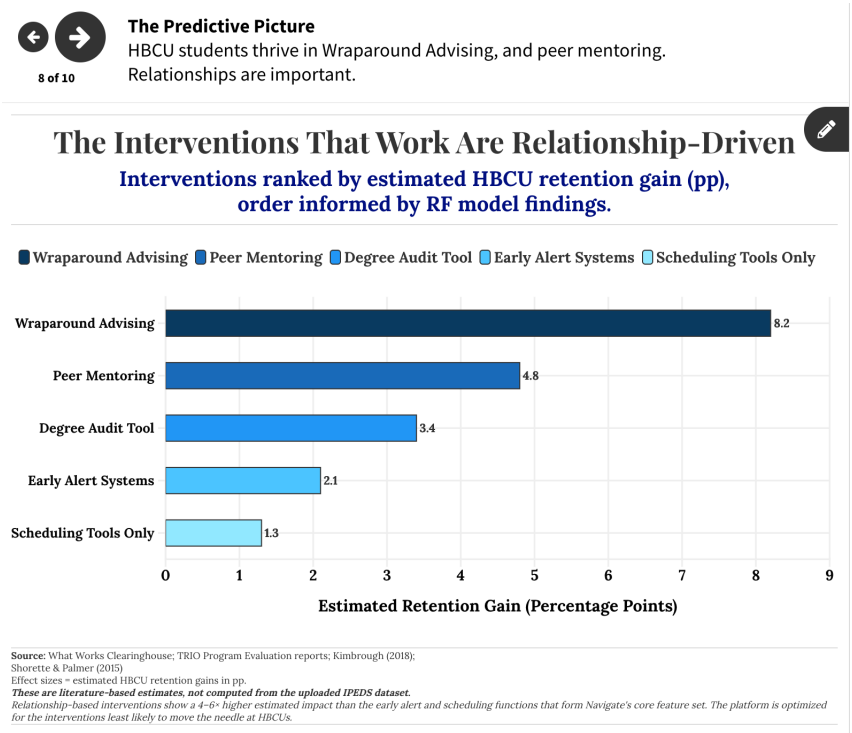
This matters because student success platforms do not operate in isolation. They require staffing, training, data infrastructure, advising capacity, intervention resources, and long-term institutional investment.

A platform may identify a risk, but the institution still has to have the resources to respond to that risk.

So when we compare HBCUs and PWIs, we cannot only ask whether both groups purchased similar tools. We also have to ask whether both groups had the same capacity to implement, customize, staff, and sustain those tools.

The answer is clearly no.

Slide 5 — The Predictive Picture Bar Chart



This slide presents the predictive results from the random forest model.

The strongest predictor in the dataset was **endowment per student**, with an importance score of **0.4571**.

By comparison, **Navigate scored only 0.0364**.

Even more striking, several major student success platform variables **scored zero**, including Early Alert, Wraparound Advising, Starfish, and Workday Student.

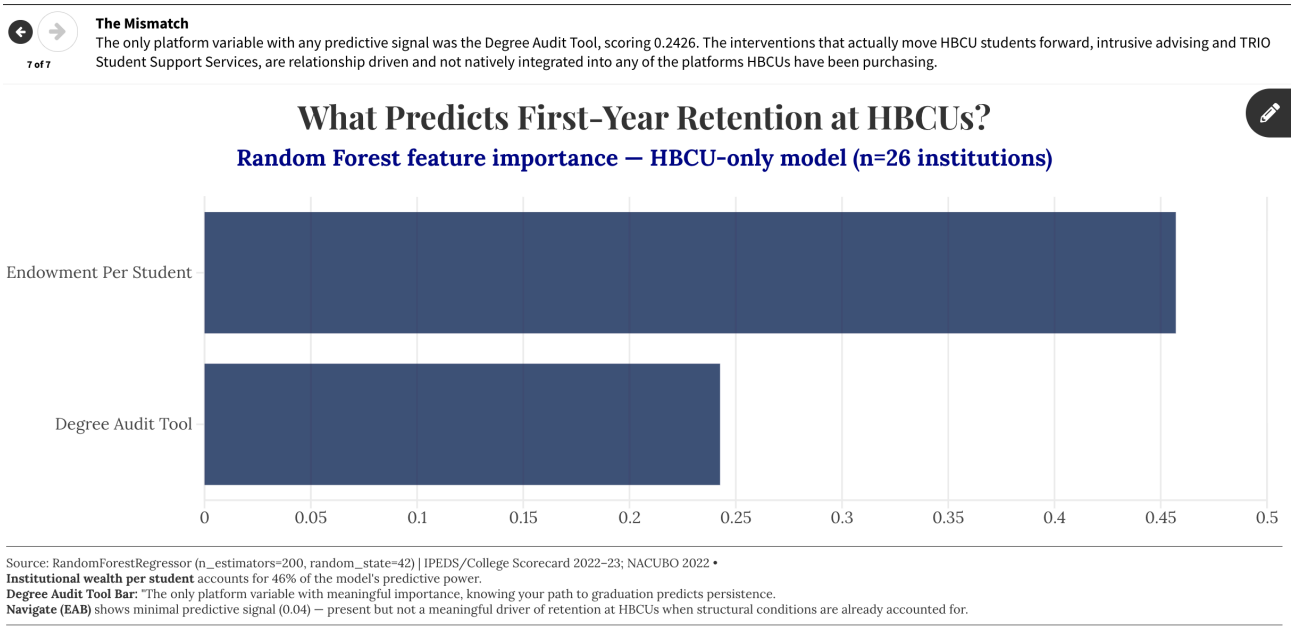
This does not mean that these tools have no value at all. But in this dataset, they did not emerge as meaningful predictors of improved retention outcomes.

The model suggests that financial capacity was far more predictive than platform adoption.

That finding is important because it challenges the assumption that purchasing a student success platform is enough to close equity gaps.

Technology may support student success, but it does not replace the institutional capacity needed to act on what the technology reveals.

Slide 6 — The Mismatch Bar Chart



This slide shows the mismatch more clearly.

The only platform-related variable with a strong predictive signal was the **Degree Audit Tool**, which scored **0.2426**.

That makes sense. Degree audit tools help students and advisors track requirements, progress, and completion pathways. They support clarity around graduation.

But the interventions that often move HBCU students forward are not just technical. They are relational.

They include **intrusive advising**, **TRIO Student Support Services**, proactive outreach, culturally affirming support, emergency aid connections, and wraparound care.

These supports depend on trust, context, and relationships.

Yet these relationship-driven interventions are not natively integrated into the major platforms HBCUs have been purchasing.

That is the mismatch.

The platforms can document risk, but they do not fully understand the lived conditions that produce the risk. And they do not always embed the culturally responsive interventions that HBCU students need.

Slide 7 — Conclusion Transition Slide

So where does this leave us?

There are currently **130,000 students** enrolled at HBCUs that adopted Navigate or comparable platforms.

And yet, the retention gap has persisted through eight years of platform adoption.

That tells us something important.

The issue is not simply whether HBCUs have technology. The issue is whether the technology fits the students, the institutions, and the conditions of HBCU student success.

The right technology has not yet been built.

And building it is not just a product enhancement.

It is an equity imperative.

If student success platforms are going to serve HBCUs effectively, they must move beyond generic risk detection and toward culturally responsive, resource-aware, relationship-centered intervention systems.

Because **when the tools do not fit, the gap does not close**.